

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Machine Learning Intern at CodTech IT Solutions Pvt Ltd”

Submitted in partial fulfillment for the award of degree(21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

E Harshith - 1DS21AI016

Conducted at



CODTECH IT SOLUTIONS PVT.LTD

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Machine Learning Intern at CodTech IT Solutions Pvt Ltd**” carried out by **Mr E Harshith[1DS21AI016]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head
Neela Santhosh Kumar
HR
CodTech IT
Solutions Pvt
Ltd

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **E Harshith**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **CODTECH IT SOLUTIONS PVT.LTD**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI016

Place: Bangalore

NAME: E Harshith

OFFER LETTER



CODTECH IT SOLUTIONS PVT.LTD IT SERVICES & IT CONSULTING

8-7-7/2, Plot NO.51, Opp: Naveena School, Hasthinapuram Central, Hyderabad , 500 079. Telangana

Internship Offer Letter



Date : 08/02/2025

Intern ID :CT4MTDN

Dear Harshith Ebudi

Congratulations on being selected for the **Machine Learning**. We at **CODTECH IT SOLUTIONS PVT.LTD** are thrilled to have you join our team. This **Online Internship** will span **4 months** from **February 10th, 2025 to June 10th, 2025**.

This internship is designed as an educational experience, focusing on learning, skill development, and gaining practical knowledge. As an intern, we expect you to:

1. Complete all assignments to the best of your ability.
2. Follow any lawful and reasonable instructions provided by your supervisors.
3. Participate actively in team meetings and discussions.
4. Provide regular updates on your progress.
5. Adhere to company policies and maintain a professional demeanor.
6. Collaborate effectively with team members and contribute to group projects.
7. Seek feedback and apply it to improve your performance.

We trust that you will approach all tasks with diligence and enthusiasm. We are confident that this internship will be an enriching experience for you. We look forward to working with you and supporting you in achieving your career aspirations

Best regards,

Neela Santhosh Kumar

Human Resources & Academic Head



VERIFIED BY



CODTECH IT SOLUTIONS PRIVATE LIMITED CODTECH IT SOLUTIONS PVT LTD

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COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Neela Santhosh Kumar**, HR&AH, CodTech IT Solutions Pvt Ltd , for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

E Harshith[1DS21AI016]

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CHAPTER 1

INTRODUCTION

In the digital age, where online platforms serve as the primary medium for public interaction, understanding audience sentiment has become vital for content creators, brands, and organizations. YouTube, being one of the largest video-sharing platforms, hosts an enormous volume of user comments that reflect real-time feedback, opinions, and emotions. Manually analyzing this data is both time-consuming and inefficient, creating the need for automated solutions.

This project focuses on building a real-time sentiment analysis system that automatically retrieves comments from a specified YouTube video and analyzes their emotional tone using advanced natural language processing techniques. The system employs a pre-trained transformer model to classify each comment as positive, neutral, or negative and assigns a sentiment score accordingly. These insights are then stored in a cloud database for persistent tracking and further analysis.

Designed for automation, the system operates continuously, updating every few minutes to capture the latest audience reactions. This allows for immediate visibility into shifts in viewer sentiment, supporting data-driven decision-making, timely content moderation, and improved audience engagement strategies.

By combining real-time data collection, sentiment evaluation, and automated reporting, the system provides a scalable and efficient tool for extracting meaningful insights from large volumes of user-generated content—enabling creators and stakeholders to stay connected with their audience like never before.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

CODTECH IT SOLUTIONS PVT LTD



Introduction

CodTech IT Solutions Private Limited is a Hyderabad-based technology services company, incorporated on December 8, 2023. Registered under the CIN U62099TS2023PTC179813, the company is headquartered in Naganool, Mahabub Nagar, Telangana. It was founded by Harish Neelam and Akhil Kumar Neela, with Harish Neelam currently serving as the CEO. CodTech IT Solutions operates as a privately held company and has a growing team of approximately 51–200 employees.

The company offers a range of IT services, including web and mobile app development, digital marketing, and IT consulting. It focuses on delivering cutting-edge solutions tailored to business needs, leveraging the latest technologies. Additionally, CodTech provides professional training and internship programs geared toward students and freshers, aiming to bridge the gap between academic learning and industry requirements. These programs often include certification, recommendation letters, and flexible work arrangements, making them attractive to early-career professionals.

CodTech IT Solutions is a certified Google Partner and AWS Partner, indicating its commitment to industry standards and technical excellence. The company maintains a strong online presence through its websites and social platforms like LinkedIn and Instagram. With its blend of service delivery and talent development, CodTech is positioning itself as a dynamic player in the Indian IT services landscape.

VISION STATEMENT

To be the trusted technology partner for small businesses, empowering them with innovative solutions in Web & App Development, AI, Data Science, and cybersecurity. The company aims to drive growth, optimize operations, and protect digital assets, enabling businesses to thrive in a digital-first world

MISSION STATEMENT

At CodTech IT Solutions, the mission is to empower small businesses with innovative Web and App Development, AI, Data Science, Cybersecurity, and Digital Marketing solutions. The goal is to help businesses optimize operations, enhance online presence, and drive growth through tailored, cutting-edge technology and marketing strategies.

CHAPTER 3

SCOPE OF WORK

During the course of my internship, I was involved in two core projects that explored cutting-edge applications of Large Language Models (LLMs) and retrieval-augmented generation (RAG) in the domain of Natural Language Processing (NLP). The work provided deep insights into language model fine-tuning, evaluation of embedding models, and optimizing LLMs for domain-specific tasks.

1.Customer Segmentation using EDA and Clustering

Objective: The primary objective of this project is to analyze customer data to uncover hidden patterns and segment customers effectively using clustering techniques. This involves two major stages: Exploratory Data Analysis (EDA) and Clustering, aimed at providing insights into customer behavior and supporting data-driven business strategies.

Scope:

- Conduct a thorough exploratory data analysis (EDA) to understand the distribution, correlation, and trends within the customer dataset.
- Perform data preprocessing, including handling missing values, feature selection, and scaling.
- Apply clustering algorithms to segment customers based on behavioral patterns.
- Evaluate and visualize the performance of clustering techniques for business insights.

Methodology:

1. Exploratory Data Analysis (EDA)

- Imported and examined the structure and summary statistics of the dataset.
- Visualized distributions using histograms, boxplots, pair plots, and heatmaps.
- Identified missing values and outliers for preprocessing.
- Used correlation analysis to identify relationships among features.

2. Data Preprocessing

- Cleaned the dataset by imputing or removing missing values.
- Standardized numerical features using scaling techniques like MinMaxScaler or StandardScaler.

- Selected relevant features for clustering based on EDA findings.

3. Clustering

- Applied unsupervised learning algorithms, particularly KMeans and Hierarchical Clustering.
- Determined the optimal number of clusters using the Elbow Method and Silhouette Score.
- Visualized clusters using scatter plots and PCA for dimensionality reduction.
- Interpreted cluster characteristics to derive customer profiles.

Results

- The EDA revealed key insights into customer behavior, spending habits, and demographic trends.
- An optimal number of customer segments was determined (e.g., 3–5 clusters).
- The clustering models successfully grouped customers with similar attributes and behaviors.
- Each cluster was profiled based on age, income, spending score, etc., enabling targeted marketing and customer relationship strategies

2. Real-Time Sentiment Analysis

Objective: The objective of this project is to develop an automated system that fetches comments from a specific YouTube video, analyzes their sentiment using a transformer-based natural language processing model, and stores the results in a cloud database (Firebase Firestore). Scope:

- **Comment Retrieval:** Collect top-level comments from a designated YouTube video using the YouTube Data API.
- **Sentiment Analysis:** Use a pre-trained transformer model (via Hugging Face) to evaluate the sentiment of each comment.
- **Data Storage:** Save each comment, along with its sentiment score, in Firebase Firestore—avoiding duplication by checking existing comment IDs.
- **Automated Processing:** Continuously run the system every 5 minutes to fetch new comments and update sentiment analysis results.
- **Sentiment Summary:** Generate and print a report with average sentiment scores and the distribution of positive, neutral, and negative comments.

Methodology:

- **Environment Setup:** Load API keys and credentials from environment variables using the `.env` file.
- **Comment Fetching:** Use HTTP requests to interact with the YouTube API, handling pagination for bulk comment retrieval.
- **Sentiment Scoring:** Apply the Hugging Face sentiment pipeline and map sentiments to numerical scores for analysis.
- **Duplicate Check:** Before storing, validate that each comment is not already present in the Firestore collection.
- **Loop Execution:** Use a timed loop (`time.sleep(300)`) to ensure the pipeline runs continuously with minimal oversight.

The system is expected to provide real-time sentiment insights by continuously analyzing new YouTube comments and storing them with sentiment scores in Firebase. Users will be able to track audience mood over time, identify shifts in sentiment, and make data-driven decisions. The automated nature of the system ensures minimal manual effort while maintaining up-to-date feedback for content moderation or engagement analysis.

CHAPTER 4

LEARNING OBJECTIVES

1. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) helps uncover patterns, detect anomalies, and build initial hypotheses, forming the basis for deeper analysis. It involves preparing data by handling missing values, outliers, and ensuring correct data types. Univariate analysis focuses on understanding individual variables through summary statistics and visualizations like histograms and boxplots. Bivariate and multivariate analyses explore relationships between variables using scatter plots and correlation matrices. Visualization tools such as Seaborn and Matplotlib support effective communication of insights. These insights lead to hypothesis generation and a clearer understanding of the dataset before modeling.

2. Customer Segmentation using Clustering Techniques

Clustering is used to group similar data points without predefined labels and is essential for applications like customer segmentation. Preprocessing steps like feature scaling ensure meaningful clustering results. The K-Means algorithm partitions data into clusters based on distance from centroids. Techniques like the Elbow Method and Silhouette Score help determine the optimal number of clusters. Profiling clusters reveals distinct customer segments, while visualization tools like PCA and t-SNE help interpret results. These insights can guide targeted marketing and customer experience strategies.

3. Sentiment Analysis from YouTube Comments using Transformers

Sentiment analysis identifies emotional tone in text data and is applied here to YouTube comments. Comments are retrieved using the YouTube Data API and analyzed using a Hugging Face transformer model. Sentiment scores classify comments as positive, neutral, or negative. Results are stored in Firestore and updated automatically to maintain real-time relevance. Sentiment summaries provide valuable feedback that can support content creation, marketing decisions, and audience engagement analysis.

4.Data Integration and Cloud Storage

Integration with cloud platforms like Firebase provides a scalable solution for managing large volumes of structured and unstructured data. In this context, Firestore is used to store YouTube comments along with their corresponding sentiment scores, enabling centralized and persistent storage. The use of the Firebase Admin SDK allows secure, authenticated access to the database and supports querying, filtering, and updating records efficiently. This setup facilitates seamless data integration into downstream processes such as dashboards, reports, or machine learning workflows, while ensuring data integrity and accessibility.

5. Automation and Scheduling of Analytical Pipelines

Automating analytical workflows enhances efficiency by eliminating the need for manual intervention in repetitive tasks. The sentiment analysis pipeline is designed to run at regular intervals, automatically fetching the latest YouTube comments, analyzing sentiment, and uploading results to Firestore. This scheduled execution ensures real-time insights and supports dynamic data environments where new information is continuously generated. Automation not only improves scalability but also supports continuous monitoring and feedback loops in production systems, making the solution robust and deployment-ready.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Handling Missing or Inconsistent Data

Challenge:

Real-world datasets often contain missing, inconsistent, or noisy values which can affect the quality of analysis and model accuracy.

Solution:

Data cleaning techniques such as imputation, removal of null entries, and normalization were applied. Careful preprocessing ensured data consistency and preserved meaningful patterns necessary for accurate clustering and EDA.

2. Selecting Relevant Features for Clustering

Challenge:

Using irrelevant or unscaled features in clustering can lead to poor segmentation and misleading results.

Solution:

Feature selection was guided by exploratory insights and domain understanding. Standardization was applied to normalize scales across features, improving the effectiveness of K-Means clustering and the interpretability of clusters.

3. Choosing the Optimal Number of Clusters

Challenge:

Arbitrarily choosing the number of clusters may result in underfitting or overfitting, affecting the usefulness of segmentation.

Solution: Techniques such as the Elbow Method and Silhouette Score were employed to evaluate different cluster counts. These metrics provided a data-driven basis for selecting the optimal number of clusters.

4. Managing API Rate Limits and Incomplete YouTube Data

Challenge:

YouTube Data API has rate limits and paginated responses, which can limit the ability to fetch all relevant comments efficiently.

Solution:

A pagination handling mechanism and retry logic were implemented to ensure complete and reliable data retrieval. Data was fetched in batches while respecting API limits, allowing full comment sections to be analyzed.

5. Ensuring Sentiment Classification Accuracy

Challenge:

General-purpose sentiment models may not always capture the context or tone specific to a niche audience or platform like YouTube.

Solution:

A pre-trained transformer model from Hugging Face was used to ensure state-of-the-art sentiment detection. Additional thresholding helped distinguish neutral sentiment, improving the classification granularity and report clarity.

6. Preventing Duplicate Storage in Firestore

Challenge:

Uploading duplicate records to Firestore can waste storage, affect performance, and skew analysis results.

Solution:

A validation step was added to check if a comment already exists in the database using its unique ID. Only new entries were uploaded, ensuring data integrity and efficient storage usage.

7. Automating Tasks Without Interruptions

Challenge:

Automated scripts running over long durations may face disruptions, memory issues, or failed API calls.

Solution:

A looping mechanism with delay intervals and exception handling was implemented to allow the script to run continuously with resilience. Failures were logged, and the script was designed to recover or retry without manual intervention.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Operating system: Windows 10 64 bit
- Python: 3.10+
- PyTorch: 2.1.2 (with CUDA support)
- Transformers (Hugging Face): 4.39.3
- Jupyter Notebook / VS Code / Terminal-based workflow

Hardware Requirements:

- Processor: Intel i5 / AMD Ryzen 5 (or equivalent high-performance CPU)
- RAM: Minimum 8 GB (for handling large datasets and model pipelines)
- GPU: NVIDIA A100 / V100 / RTX 3050
- Input Devices: Mouse, Keyboard
- LAN Connection Port: Required
- Internet Fiber Optic Cable: Required

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Real Time Sentiment Analysis

- **End-to-End Pipeline Development:** Successfully built a complete data pipeline integrating data extraction, preprocessing, modeling, analysis, and storage. This pipeline efficiently handled real-time YouTube comments and enabled continuous sentiment monitoring.
- **Implementation of Advanced Analytical Techniques:** Applied Exploratory Data Analysis (EDA) techniques to derive meaningful insights and used K-Means clustering to segment data into logical groupings. These methods allowed for the identification of patterns and trends that would otherwise remain hidden.
- **Integration of NLP with Real-Time Data:** Utilized a pre-trained transformer model for high-accuracy sentiment analysis, demonstrating the ability to apply modern NLP techniques to social media data. This added qualitative depth to the project beyond numeric or categorical analysis.
- **Cloud-Based Storage and Automation:** Integrated Firebase Firestore for persistent and scalable data storage, ensuring that sentiment scores and comments are safely archived. Additionally, the process was automated to run at defined intervals, enabling continuous data ingestion and analysis.
- **Scalable and Modular System Design:** The system was designed to be modular, allowing easy extension or modification. Each component—data fetching, processing, sentiment analysis, clustering, and storage—can function independently or be upgraded without impacting the whole pipeline.
- **Technical Design and Implementation:** Designed and coded the entire architecture of the project including API integration, NLP model usage, clustering logic, and cloud connectivity. Emphasis was placed on maintainability and clarity.
- **Data Analysis and Insight Generation:** Conducted EDA and clustering to extract actionable insights from the data. This included interpreting cluster characteristics and sentiment distributions to support use cases such as customer segmentation or content strategy.
- **Automation and Scheduling:** Implemented a looping mechanism and integrated time delays to ensure the system could fetch, analyze, and upload new data in regular intervals without manual input, simulating a real-time production system.

- **Documentation and Reporting:** Created thorough documentation of the process, including code comments, architecture flow, and analytical findings. This improves reproducibility and facilitates handoff or scaling of the system in future development.

CHAPTER 7

REFLECTION AND ANALYSIS

This project offered a comprehensive exploration of real-world data analysis workflows, combining exploratory data analysis, unsupervised machine learning, natural language processing, and cloud-based automation. The process not only deepened technical proficiency but also provided valuable insights into the practical challenges faced when working with large-scale, dynamic datasets. One of the most significant takeaways was the importance of data quality and preprocessing. Initial efforts in cleaning and preparing the data laid the foundation for meaningful analysis. Without careful attention to missing values, outliers, and inconsistent formats, later stages such as clustering and sentiment analysis would have produced unreliable results. This reinforced the value of data wrangling as a crucial step in any analytical pipeline.

The implementation of K-Means clustering highlighted the strengths and limitations of unsupervised learning. It required thoughtful feature selection and a deep understanding of the data to ensure the clusters were interpretable and actionable. The process of selecting the optimal number of clusters using the Elbow Method and Silhouette Score offered valuable experience in model evaluation, a skill transferable to many areas of machine learning. Incorporating sentiment analysis using a transformer-based model expanded understanding of natural language processing. Working with YouTube comments introduced challenges such as slang, sarcasm, and mixed sentiments, which demanded more robust tools than basic text classification. Leveraging Hugging Face's sentiment pipeline provided both simplicity and high performance, demonstrating the power of pre-trained models in applied NLP tasks.

The integration with Firebase Firestore introduced practical experience with cloud services and data persistence. It emphasized the importance of scalable, real-time databases in applications where data is continuously generated. Designing the pipeline to prevent duplicate storage also encouraged careful database design and efficiency considerations. Automating the entire process reinforced the significance of reliability and system resilience in production-like environments. Building a looped, error-tolerant system that could operate without manual input simulated real-world data engineering challenges and provided a sense of how data-driven applications are maintained in the industry.

Overall, the project bridged multiple disciplines data preprocessing, machine learning, sentiment analysis, and cloud engineering into a cohesive system. It offered not just technical growth but also a deeper appreciation for how different components come together to form complete, scalable analytical solutions. This holistic understanding is invaluable for tackling future data science problems in both academic and professional contexts.

CHAPTER 8

RECOMMENDATIONS

- **Fine-tune the sentiment model:** While the pre-trained transformer model performs well, training it further on YouTube-specific comment data can improve accuracy. This helps capture informal language, sarcasm, and platform-specific context more effectively, leading to more reliable sentiment classification.
- **Expand data collection:** Including additional data like comment replies, timestamps, and user metadata enables deeper analysis. This allows for sentiment trends over time, user behavior analysis, and better segmentation for marketing or content strategy.
- **Use advanced clustering techniques:** Beyond K-Means, algorithms like DBSCAN or Gaussian Mixture Models can uncover more complex cluster structures. These methods are particularly useful when dealing with overlapping groups or irregular data distributions.
- **Implement interactive visualizations:** Tools like Streamlit or Dash allow users to interact with charts and filters in real time. This enhances user experience by enabling deeper exploration of data without requiring technical expertise.
- **Containerize and deploy on the cloud:** Packaging the application with Docker and deploying it on platforms like Google Cloud or AWS improves scalability. This setup supports automated scheduling, reduces downtime, and simplifies deployment across environments.
- **Optimize Firestore structure:** Structuring Firestore collections with proper indexing and subcollections improves query performance. Organizing data by topic or time also simplifies analytics and supports faster data retrieval.
- **Enable result versioning and history tracking:** Keeping a version history of sentiment results helps track changes over time and provides auditability. This is useful for monitoring trends and reverting to previous data if needed.

CHAPTER 9

CONCLUSION

This project successfully demonstrated the end-to-end integration of data analysis, machine learning, and cloud-based automation to derive insights from user-generated content. Starting with exploratory data analysis, meaningful patterns and trends were uncovered that informed the structure and direction of the clustering model. Through customer segmentation using K-Means, the project highlighted how businesses can extract value by identifying distinct groups within a dataset, enabling more targeted decision-making.

The application of sentiment analysis to YouTube comments introduced a powerful layer of natural language processing, providing a deeper understanding of user engagement and public opinion. Leveraging transformer models from Hugging Face ensured state-of-the-art sentiment classification, while real-time data fetching through the YouTube Data API made the analysis dynamic and continuously relevant.

By storing results in Firebase Firestore, the system gained persistent and scalable data management capabilities, making it suitable for live environments. The addition of automation and scheduling transformed the project into a production-ready pipeline capable of functioning with minimal human intervention. These features demonstrate the potential of combining data science techniques with software engineering practices to build intelligent, self-sustaining systems.

In addition to technical skills, this project emphasized problem-solving, adaptability, and the importance of planning scalable architectures. Challenges such as data inconsistencies, API limitations, and clustering decisions were tackled through a combination of domain understanding and iterative refinement. The experience gained offers a solid foundation for future projects that involve real-time data analytics, machine learning, and cloud computing.

Overall, the project reflects a well-rounded application of data science concepts to solve practical problems. It serves not only as a technical showcase but also as a blueprint for building modular, scalable, and intelligent analytical systems that can adapt and evolve with real-world data.

CHAPTER 10

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